

GREAT LAKES INSTITUTE OF MANAGEMENT, GURGAON

Post Graduate Certificate in Management (PGPM 2026-27)

Term II - Operations Management Course Project

PARCEL MANAGER

Parcel Distribution and Operation System

A Simple Operations Management App with a 3D Dashboard for Campus Parcel Handling

Project Team Members

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Abstract

Campus students receive many online parcels every day. This creates problems like crowded mailrooms, late alerts, full storage racks and long waiting lines. Our team built a simple web app called “Parcel Manager” to fix these issues.

The app uses basic operations management ideas such as tracking process time, checking service levels, showing storage heatmaps, 3D views of racks, waste checks and staffing plans. We tested it with data from 2,500 parcels, 500 students and 1,200 storage slots.

Results: 83.6% of parcels met the time target and the system gives automatic tips to balance full storage racks.

1. Introduction

At Great Lakes Institute of Management, Gurgaon, more than 500 students live on campus. Each student gets about 5 parcels per term from sites like Amazon, Flipkart, Myntra and others. On busy days, over 500 parcels arrive.

In the past, staff used registers. This caused delays at the gate, long student queues and crowded shelves. A digital app makes the work faster, clearer and easier to manage with data.

2. Problem Statement

The mailroom had four main problems:

- Slow processing: Average handling time was 55.1 minutes. Almost half of this time (27.5 minutes) was spent waiting.
- Uncollected parcels: 17% of parcels (425 items) stayed more than 7 days. This filled up shelf space.
- Uneven storage use: Some racks were 82% full while nearby ones were only 35% full.
- Errors: Manual lookup caused a 20% defect rate (missing, unclaimed or damaged parcels).

3. Project Objectives

1. Build a clear, easy-to-use web app that matches Great Lakes branding.
2. Give staff simple forms to log and move parcels in real time.
3. Show storage levels with 2D heatmaps and interactive 3D models.
4. Create automatic alerts (Critical, High, Medium) for capacity and staffing needs.
5. Provide students with QR codes and 4-digit OTP codes for quick pickup.

4. Scope of the Project

In Scope:

Receiving parcels at the gate, assigning storage lockers, student pickup checks, live filters, capacity views (2D and 3D), time tracking, demand forecasts and decision alerts.

Out of Scope:

Tracking parcels before they reach campus and managing courier billing.

5. Operations Management Concepts Applied

5.1 Process Time Breakdown

Total internal handling time of 55.1 minutes was split into three parts:

1. Registration Time (15.2 mins): Unloading and entering barcode at the gate.
2. Putaway Time (12.4 mins): Moving the parcel from gate to storage rack.
3. Notification Lag (27.5 mins): Delay before sending the SMS alert to the student.

Target: Finish within 60 minutes. Current success rate: 83.6%.

5.2 Storage Capacity Planning

The campus has 6 main racks in 3 rooms with 1,200 locker slots in total. Utilization is calculated as $(\text{occupied slots} \div \text{total slots}) \times 100$. The 3D view colors racks green (under 50%), amber (50 – 75%), and red (75% or more).

5.3 Lean Waste Check (8 Wastes)

We used Lean ideas to find and reduce waste:

Waste Type	Count	% of Total	Suggested Fix
Inventory (overdue parcels)	425	17.0%	Return policy after 5 days
Defects (missing parcels)	275	11.0%	QR code + 4-digit OTP checkout
Waiting (student queues)	190	7.6%	Express OTP counter + extra staff
Motion (searching racks)	110	4.4%	Clear labels on every compartment

5.4 Demand Forecast and Staffing

Most parcels arrive between 4 pm and 6 pm (42.8% of the day). Using a simple queue model, we suggest 2 staff during quiet hours and 4 staff during peak hours to avoid bottlenecks at the gate.

6. Dashboard Features

The app has seven main tabs:

1. Executive Overview - Key numbers (total parcels, lead time, SLA %, storage use, defect rate) and charts.
2. Inbound Parcel Manager - Form for staff to receive and log parcels, with student list and live filters.
3. Rack Capacity Heatmap - Simple 2D bars showing how full each of the 6 racks is.
4. 3D Rack Visualizer - Interactive 3D model of the mailroom that you can rotate, zoom, and click.
5. Decision Support & Alerts - Automatic priority messages (Critical, High, Medium) and audit log.
6. Lean & Waste Analysis - Pareto chart of defects and the 8-waste summary table.
7. Demand & Staffing Model - Chart showing hourly arrivals next to recommended staff numbers.

7. Sample Data and Database

The system uses an Excel workbook with 10 sheets. Main ones include:

- Parcels_Master - 2,500 parcel records with tracking numbers and times.
- Storage_Locations - 6 racks across 3 rooms with capacity details.
- Recipients_Master - 504 people (500 students + 4 staff).
- Other sheets for KPIs, waste analysis, alerts and demand forecasts.

8. Results and Findings

- SLA Compliance: 83.6% of parcels finished within the 60-minute target.
- Faster Alerts: Automatic SMS at putaway removed the 27.5 minute notification lag.
- Better Storage Use: System recommends moving parcels from full racks (like SL-B1) to emptier ones (like SL-A1).
- Fewer Errors: QR code and OTP checks cut missing-parcel mistakes by about 80%.

9. Challenges Faced and Solutions

- Browser kept showing the old logo. Fixed by adding a version number to the image link so the browser loads the new file.
- 3D view did not update when data changed. Fixed by linking the 3D colours and counts directly to the live storage data.

10. Future Enhancements

- Connect to smart electronic lockers so students can pick up parcels themselves.
- Add camera scanning (OCR) to read courier labels automatically at the gate.
- Send mobile push notifications through a progressive web app.

11. Conclusion

The Parcel Manager system shows how simple operations management ideas and a no-code web app can solve real campus logistics problems. By turning manual parcel handling into a clear, data-driven process with 3D views and automatic alerts, Great Lakes Gurgaon can reduce waiting times, use storage better, and give students a smoother experience.

12. References

- Krajewski, L. J., Malhotra, M. K., & Ritzman, L. P. (2019). Operations Management: Processes and Supply Chains (12th ed.). Pearson.
- Stevenson, W. J. (2021). Operations Management (14th ed.). McGraw-Hill Education.
- Womack, J. P., & Jones, D. T. (2003). Lean Thinking. Free Press.
- Heizer, J., Render, B., & Munson, C. (2020). Operations Management (13th ed.). Pearson.
- Anupindi, R., et al. (2012). Managing Business Process Flows (3rd ed.). Pearson.